

Few subsequence sums in $\mathbb{Z}_n \times \mathbb{Z}_n$

OPG-156

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Conjecture For every $0 \leq t \leq n - 1$, the sequence in $\mathbb{Z}_n^2$ consisting of $n - 1$ copies of $(1, 0)$ and t copies of $(0, 1)$ has the fewest number of distinct subsequence sums over all zero-free sequences from $\mathbb{Z}_n^2$ of length $n - 1 + t$.

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>
Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>